Predicting Student Academic Performance Using Machine Learning Algorithms:
A Comparative Study on the StudentsPerformance Dataset

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Abstract

Academic performance prediction is a vital application of machine learning in educational data mining, enabling early intervention for at-risk students and supporting data-driven institutional policies. This paper presents a comprehensive comparative analysis of several supervised machine learning algorithms applied to the StudentsPerformance dataset, which contains 1000 records of student demographics and exam scores in mathematics, reading, and writing. The study implements K-Nearest Neighbors (KNN) with hyperparameter tuning via GridSearchCV, Logistic Regression, Decision Tree Classifier, Random Forest Classifier, and Support Vector Machine (SVM). The methodology follows a structured pipeline: data loading and exploration, feature engineering to create an average score and binary pass/fail target ('result' column where average ≥ 50 indicates Pass), label encoding, train-test splitting (80/20), and feature scaling using StandardScaler. Models were evaluated using accuracy and confusion matrices, with 5-fold stratified cross-validation for robust performance estimation. Results demonstrated exceptionally high predictive power across models, with tuned KNN, Logistic Regression, and SVM achieving 100% test accuracy, while Decision Tree and Random Forest also performed near-perfectly (98–99.5%). Explainable AI aspects, including feature importance from Random Forest and decision paths from Decision Trees, highlight the dominant role of subject scores in predictions.

I. INTRODUCTION

In the era of big data and artificial intelligence, educational institutions increasingly leverage machine learning to analyze student performance patterns, predict outcomes, and design personalized interventions. Accurate prediction of academic success can help identify students who may struggle, optimize resource allocation, and improve overall learning outcomes. Traditional methods relying solely on manual assessment often fail to capture complex interactions between demographic factors and academic achievement. Machine learning offers a scalable, data-driven alternative capable of modeling these relationships effectively.

This paper focuses on binary classification of student performance (Pass/Fail) using the publicly available StudentsPerformance dataset. The dataset includes 1000 records with categorical features (gender, race/ethnicity, parental level of education, lunch type, test preparation course) and continuous features (math, reading, and writing scores). An average score is engineered, and a binary 'result' label is derived (Pass if average ≥ 50). Five prominent algorithms — KNN, Logistic Regression, Decision Tree, Random Forest, and SVM — are implemented and rigorously compared.

The study adheres to best practices in reproducible research, incorporating stratified cross-validation and detailed performance metrics. By integrating explainable AI techniques such as feature importance analysis, this work addresses the "black-box" concern in educational AI applications, ensuring stakeholders can trust and act upon model insights.

The remainder of this paper is organized as follows: Section II reviews relevant literature; Section III details the methodology; Section IV presents results and discussion; Section V explores explainable AI aspects; and Section VI concludes with future research directions.

II. LITERATURE REVIEW

Educational data mining (EDM) has emerged as a significant subdomain of data science, with numerous studies applying machine learning to predict student outcomes. Cortez and Silva (2008) pioneered the use of decision trees and neural networks on Portuguese student performance data, achieving promising results with socioeconomic and behavioral features [1]. Subsequent works expanded this to diverse datasets, often reporting accuracies above 80–90% for binary pass/fail tasks.

KNN has been frequently employed in EDM due to its simplicity and instance-based learning nature. Studies by Bunkar et al. (2012) demonstrated KNN's effectiveness in classifying student performance but noted sensitivity to feature scaling and the choice of k [2]. Logistic Regression serves as a strong interpretable baseline, revealing feature influence through model coefficients — for instance, the impact of test preparation on outcomes.

Ensemble methods such as Random Forest have shown superior robustness, as highlighted in works by Romero et al. (2013), where they outperformed single trees by reducing prediction variance [3]. SVM, particularly with linear or RBF kernels, excels in high-dimensional feature spaces common in educational datasets.

Recent research emphasizes explainability. Tools like SHAP and feature importance from tree-based models help educators understand why a student is predicted to fail. This study builds upon these foundations by applying a complete pipeline — including preprocessing, scaling, tuning, and cross-validation — to the StudentsPerformance dataset.

III. METHODOLOGY

A. Dataset Description

The StudentsPerformance dataset comprises 1000 records with eight features: gender (binary), race/ethnicity (five groups), parental level of education (six categories), lunch (standard/free-reduced), test preparation course (completed/none), and three continuous exam scores (math, reading, writing; ranging 0–100). No missing values were present. The dataset is well-suited for binary classification after engineering the target variable.

B. Data Preprocessing and Feature Engineering

Data was loaded using pandas. Exploratory analysis confirmed structure and data types. A new 'average' feature was computed as the arithmetic mean of the three exam scores. The binary target 'result' was derived: 'Pass' if average ≥ 50, otherwise 'Fail'. Features for modeling (X) comprised the three scores; the target (y) was label-encoded using LabelEncoder. Data was split into 80% training and 20% testing sets. Feature scaling was applied using StandardScaler to normalize distributions — critical for distance-based algorithms such as KNN.

C. Model Implementation

The following five supervised classifiers were implemented:

- KNN — Instance-based classifier. Hyperparameter tuning via GridSearchCV explored n_neighbors (1–20) and distance metrics (Euclidean, Manhattan, Minkowski). Best parameters: n_neighbors=13, metric=Euclidean.
- Logistic Regression — Linear probabilistic model using solver='liblinear'.
- Decision Tree — CART algorithm with default parameters; provides interpretable decision rules.
- Random Forest — Ensemble of 100 trees; reduces overfitting through bagging.
- SVM — Linear kernel for baseline comparison in separable feature space.

All models were trained on scaled features where appropriate, ensuring fair comparison.

D. Evaluation Metrics

Primary evaluation metric: classification accuracy. Additional metrics included confusion matrices and per-class precision/recall. Model robustness was assessed via 5-fold stratified cross-validation, reporting mean accuracy and standard deviation across folds.

IV. RESULTS AND DISCUSSION

Exploratory analysis confirmed a clean dataset with strong correlations between exam scores and the derived 'result' label. Table I summarizes the performance of all models.

TABLE I — Comparative Model Performance Summary

Algorithm	Test Acc. (%)	CV Acc. (%)	CV Std Dev	Remarks
Tuned KNN	100.00	99.25	±0.004	Best overall stability
Logistic Regression	100.00	99.50	±0.003	Strong linear baseline
Decision Tree	98.00	97.75	±0.006	Interpretable rules
Random Forest	99.50	99.00	±0.005	Robust ensemble
SVM (Linear)	100.00	99.25	±0.004	Excellent generalization

Tuned KNN achieved 100% test accuracy and 99.25% cross-validation accuracy, demonstrating high stability following feature scaling. Logistic Regression and SVM also reached perfect test accuracy (100%), confirming linear separability in the three-dimensional score feature space. The Decision Tree yielded 98% accuracy with minor misclassifications at score boundaries, while Random Forest achieved 99.5%, reflecting the variance-reduction benefit of ensemble learning.

Cross-validation confirmed low variance across all models, with standard deviations under 0.006. Feature scaling was pivotal for KNN — without normalization, Euclidean distances are dominated by high-range score differences, leading to suboptimal neighbor selection.

The near-perfect results across all models reflect the relatively straightforward nature of this binary classification task: the derived average score is directly linked to the Pass/Fail label with a threshold of 50. Real-world extensions involving noisier or higher-dimensional data would benefit from regularization and more complex feature engineering.

V. EXPLAINABLE AI ANALYSIS

Interpretability is critical in educational AI applications, where predictions must be trusted by administrators, instructors, and counselors. This section presents explainability insights derived from two model families:

Random Forest Feature Importance: Scores for mathematics, reading, and writing were uniformly identified as the most important predictive features, with math scores consistently ranking highest in feature importance rankings. This aligns with domain knowledge about the role of quantitative skills in overall academic assessment.

Decision Tree Visualization: Decision paths from the trained CART tree reveal clear, human-interpretable thresholds. For example, a math score greater than 45 strongly predicts a Pass outcome. Such visualizations allow educators to pinpoint subject-specific intervention targets.

These XAI elements transform black-box statistical predictions into actionable intelligence. By surfacing the reasoning behind each classification, stakeholders can act upon model insights with confidence, addressing the ethical and practical concerns associated with AI-driven academic decisions.

VI. CONCLUSION

This study demonstrates the high efficacy of supervised machine learning for student academic performance prediction, with multiple models achieving near-perfect classification accuracy on the StudentsPerformance dataset. The structured pipeline — encompassing feature engineering, preprocessing, hyperparameter tuning, and stratified cross-validation — provides a reproducible and rigorous framework for educational data mining.

KNN (tuned), Logistic Regression, and SVM achieved 100% test accuracy, while Decision Tree and Random Forest followed with 98% and 99.5% respectively. Explainable AI techniques further demonstrated that subject exam scores — particularly mathematics — are the primary predictors of Pass/Fail outcomes.

Future research directions include: (1) multi-class grade prediction (A/B/C/D/F); (2) integration of additional socio-economic and behavioral features; (3) deployment in real-time student monitoring dashboards; and (4) application to larger, longitudinal datasets with temporal learning patterns. These advances would further bridge the gap between academic ML research and actionable educational practice.

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